

I. INTRODUCTION: WHY LINEAR ALGEBRA?

In the world of Machine Learning and Deep Learning, data is not processed as individual numbers but as structured collections called **Tensors**. Linear Algebra provides the language and the operations to transform these tensors. Whether you are rotating a 3D object in Computer Vision or calculating the importance of a word in NLP, you are performing Linear Algebra.

1.1. Data Representation

- **Scalar (0D):** A single number (e.g., the age of a student).
- **Vector (1D):** An ordered list of numbers (e.g., [Age, Grade, Attendance]).
- **Matrix (2D):** A table of numbers (e.g., a grayscale image where each pixel is an entry).
- **Tensor (nD):** A multi-dimensional array (e.g., a color image with RGB channels or a video sequence).

II. KEY MATRIX OPERATIONS IN MODEL TRAINING

2.1. Matrix Multiplication (Dot Product)

The most critical operation in any Neural Network. When an input (x) passes through a hidden layer, it is multiplied by a weight matrix (W).

$$y = f(W \cdot x + b)$$

Without high-speed matrix multiplication (optimized by NumPy or GPUs), training a model like GPT would take centuries instead of weeks.

2.2. Matrix Transposition (A^T)

Transposition is the process of flipping a matrix over its diagonal. In Machine Learning, this is essential during **Backpropagation**. To calculate how much to change a weight, we must align the dimensions of the error gradient with the weight matrix.

2.3. Identity and Inverse Matrices

An Inverse Matrix (A^{-1}) is used to solve systems of linear equations. In optimization, finding the inverse of a matrix (or an approximation) helps in reaching the "Global Minimum" of a loss function more efficiently.

III. ADVANCED CONCEPTS FOR DIMENSIONALITY REDUCTION

3.1. Eigenvalues and Eigenvectors

An eigenvector is a vector whose direction remains unchanged when a linear transformation is applied to it.

- **Application: Principal Component Analysis (PCA).** In Big Data, we often have thousands of features (columns). PCA uses eigenvalues to identify which "directions" in the data contain the most information, allowing us to compress 1,000 columns into 50 without losing significant accuracy.

3.2. Singular Value Decomposition (SVD)

SVD is the foundation of modern **Recommendation Systems** (like Netflix or Spotify). By decomposing a massive "User-Item" matrix, SVD can predict what a user might like based on latent features.

IV. COMPUTATIONAL IMPLEMENTATION WITH PYTHON

While mathematicians do this by hand, engineers use libraries. In your DTU lab sessions, you will use:

Python

```
import numpy as np
```

```
# Creating a 2x3 Matrix
```

```
A = np.array([[1, 2, 3], [4, 5, 6]])
```

```
# Matrix Multiplication
```

```
B = np.random.rand(3, 2)
```

```
C = np.dot(A, B)
```

```
# Calculating Eigenvalues
```

```
eigenvalues, eigenvectors = np.linalg.eig(np.eye(3))
```